



SCIENCE FOUNDATIONS

Data Analysis

Understanding your results by turning data into a graph.

PART 1

Why Graphs?



Goals for Today

1

Make a Graph

From data table → graph.

Labels, axes, type of graph, and how to plot correctly.

2

Describe a Graph

Learn the vocabulary:

- linear
- non-linear
- exponential

3

Interpret a Graph

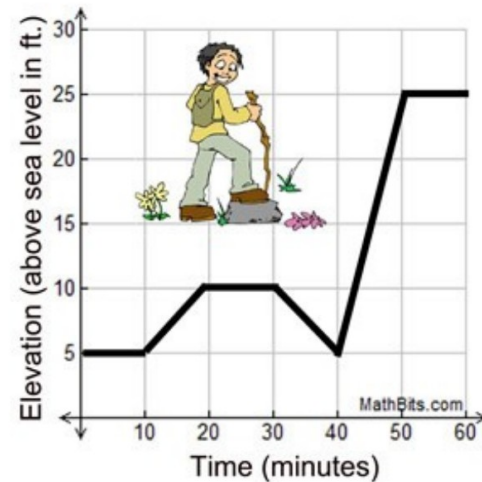
Find the RELATIONSHIP:

- proportional / inversely proportional
- positive / negative
- what the SLOPE means

Why Graphs?

Why Use Graphs

- Organizing our data into a graph does more than make it aesthetically pleasing (pretty)
- It tells a story



Start with a Data Table

MATH

generic

X	Y
0	0
1	2
2	4
3	6

GENERAL

add roles

IV	DV
0	0
1	2
2	4
3	6

SPECIFIC

add units

Time (s)	Distance (m)
0	0
1	2
2	4
3	6

SCIENCE

compact

t (s)	d (m)
0	0
1	2
2	4
3	6

← GENERAL labels get more descriptive as you move right SPECIFIC →

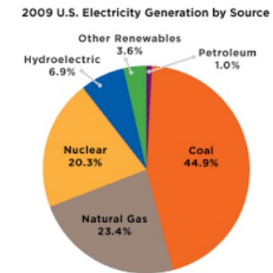
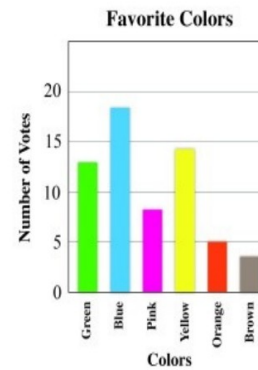
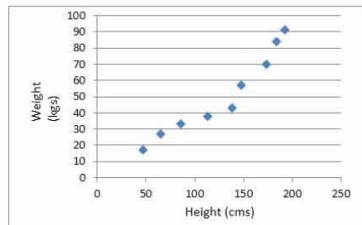
PART 2

Types of Graphs

Three Common Graph Types

Graph Types

- X-Y Scatter Plot
- Bar/Column Graph
- Pie Chart



X-Y Scatter Plot



When to Use

To find a RELATIONSHIP between two continuous variables.

- X-axis → Independent Variable (what you change)
- Y-axis → Dependent Variable (what you measure)

Most of your science-lab graphs will be this type.



Rules

- PLOT the dots, don't connect them.
- If the pattern is linear or exponential, add a TRENDLINE — the average of your data.
- Never draw a zig-zag from dot to dot; that hides the trend.

Bar/Column and Pie Charts



Bar / Column

Compare two or more different groups.

They do NOT show change over time. No relationship.

Bar = horizontal, Column = vertical. Length of the bar = the value it represents.



Pie Chart

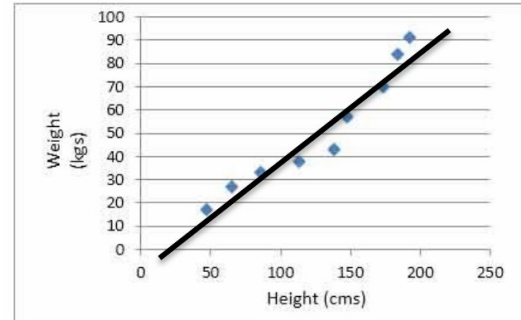
Compare parts of a WHOLE.

They do NOT show change over time. No relationship.

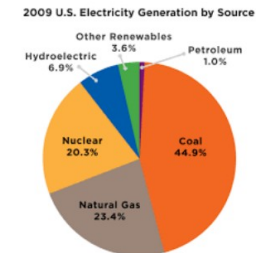
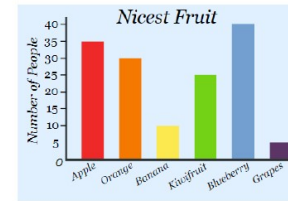
Each slice = its share of 100%.

Where Is the Relationship Shown?

Where is the Relationship Shown?



RELATIONSHIP BETWEEN Y-
VALUE AND X-VALUE



NO RELATIONSHIP

PART 3

Interpreting a Scatter Plot

Three Questions to Ask

1

Shape

LINEAR or NON-LINEAR?

2

Direction

POSITIVE or NEGATIVE relationship?

3

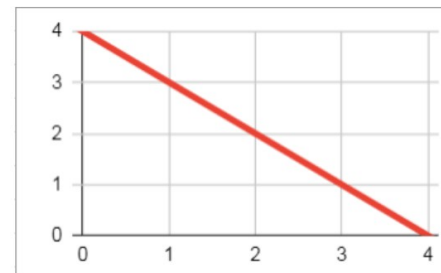
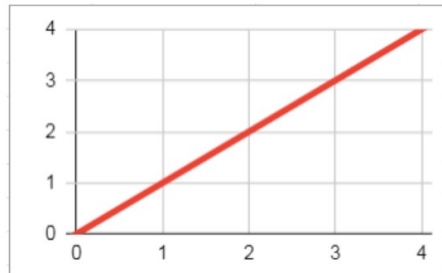
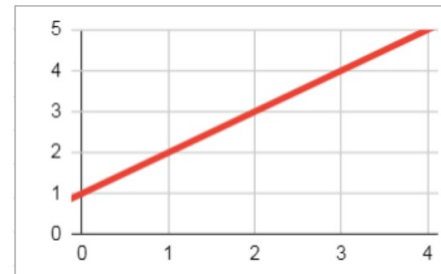
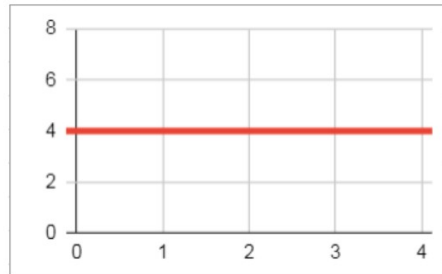
Proportion

PROPORTIONAL, DIRECTLY
PROPORTIONAL, or INVERSELY
PROPORTIONAL?

Linear vs. Non-Linear

Linear & Constant Slope

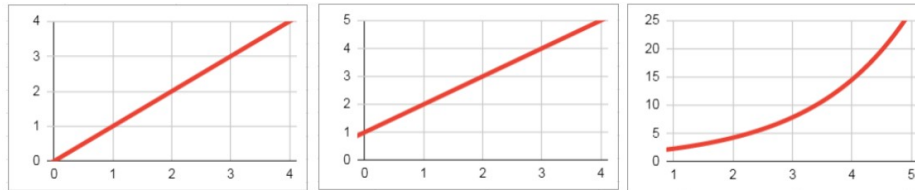
- These types of graphs will always have straight lines



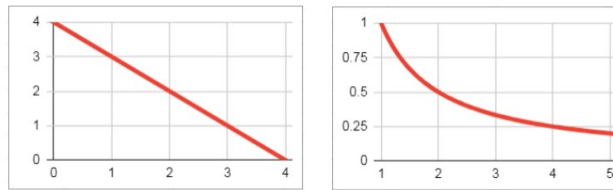
Positive vs. Negative Relationship

Positive or Negative Relationship

- Positive - as x-axis increases, y-axis increases too



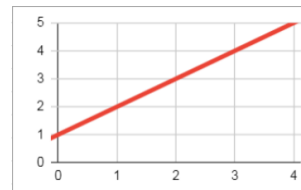
- Negative -as x-axis increases, y-axis decreases



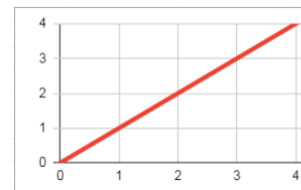
Proportionality

Proportionality

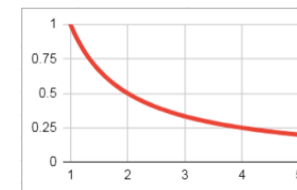
- Proportional – as IV increases, DV increases
- Directly Proportional – as IV increases, DV increases and goes through 0,0
- Inversely Proportional – as IV increases, DV decreases



PROPORTIONAL



DIRECTLY
PROPORTIONAL



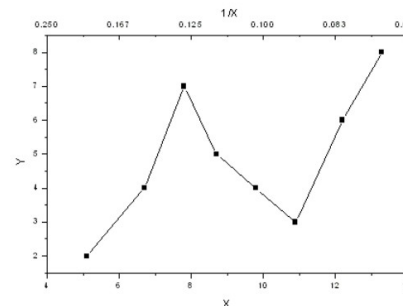
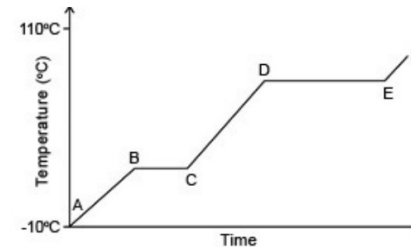
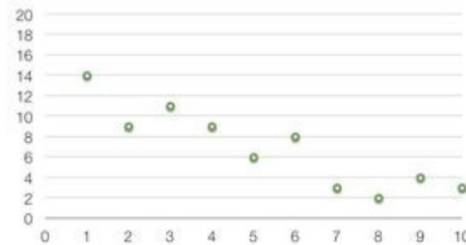
INVERSELY
PROPORTIONAL

Non-Linear Patterns

Nonlinear

- Not a straight line path
- Look at the WHOLE picture for trends or patterns
 - may or may not be present

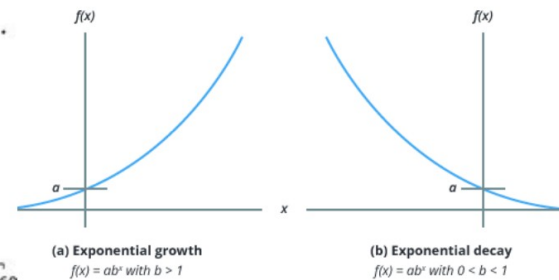
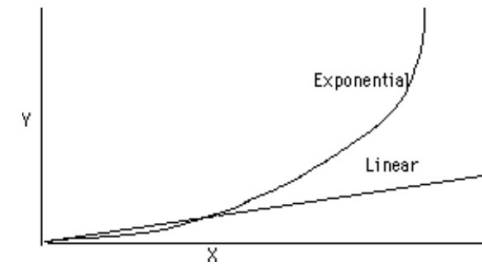
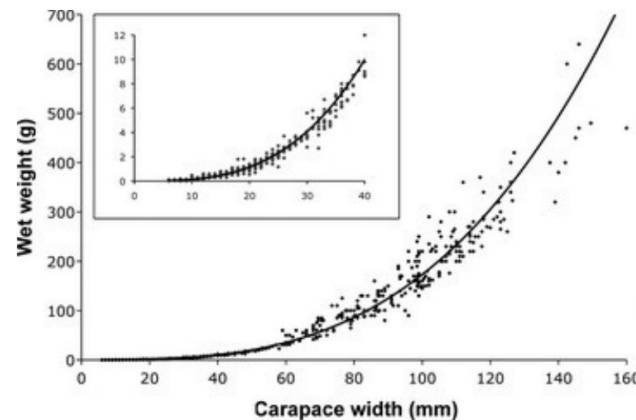
Example 3. Non-Linear Graph



Exponential

Exponential

- Graph increases or decreases by huge factors (shaped as a curve)



PART 4

Practice — Read These Graphs



For Each Graph, Answer:

1

Shape?

Linear / non-linear / exponential?

2

Direction?

Positive or negative?

3

Proportion?

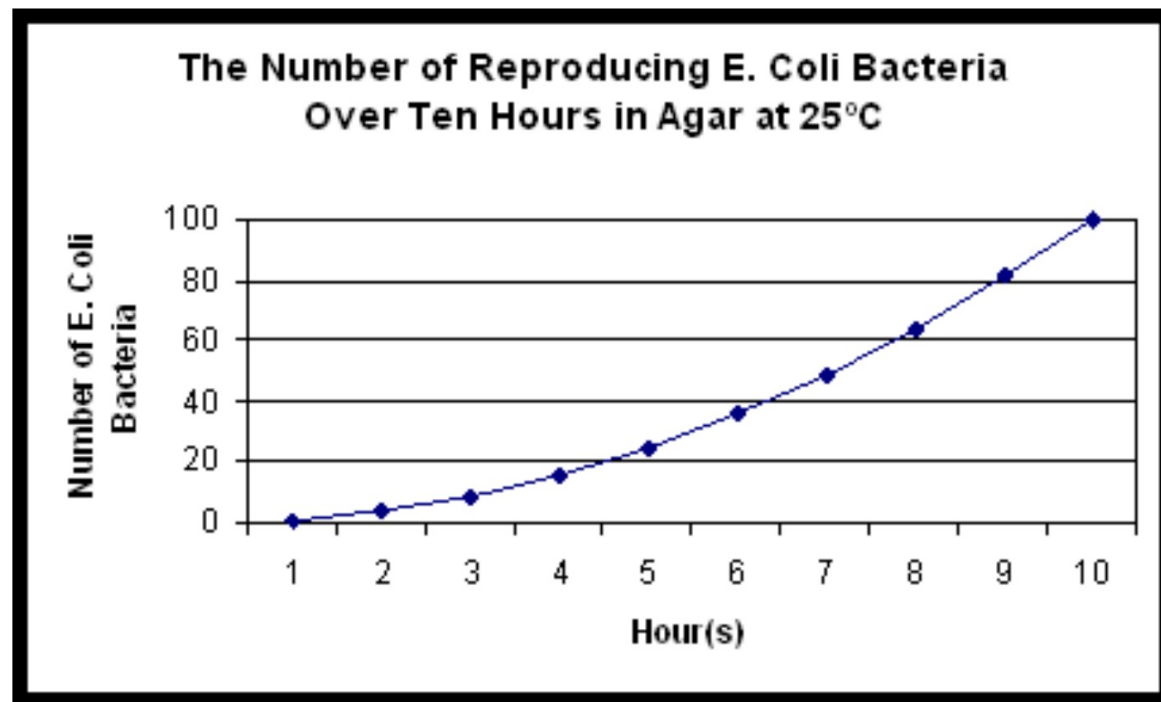
Proportional, directly proportional, or inversely?

4

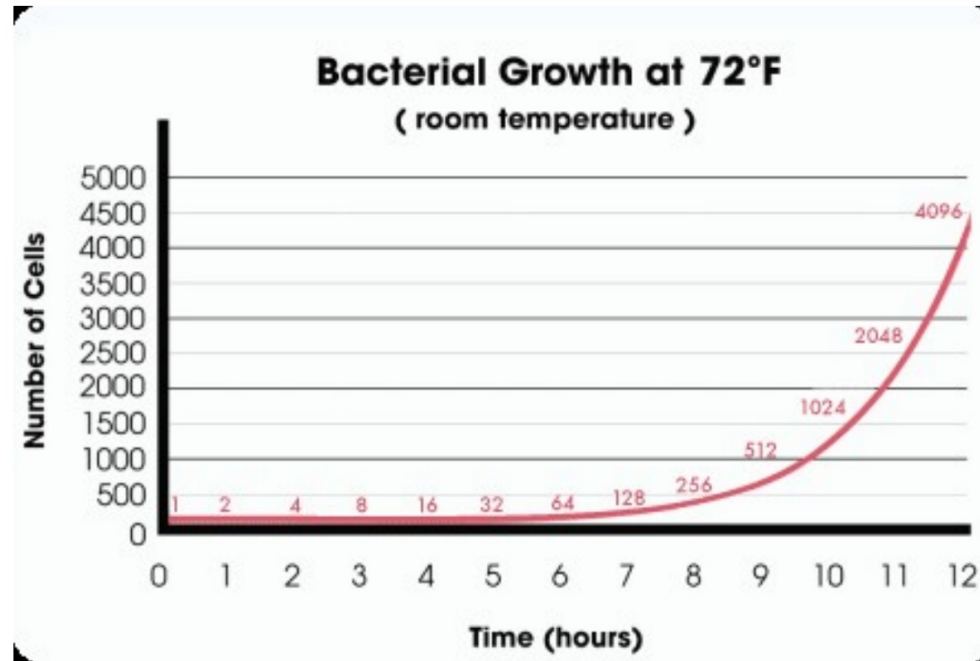
Story?

'As X increases, Y _____'

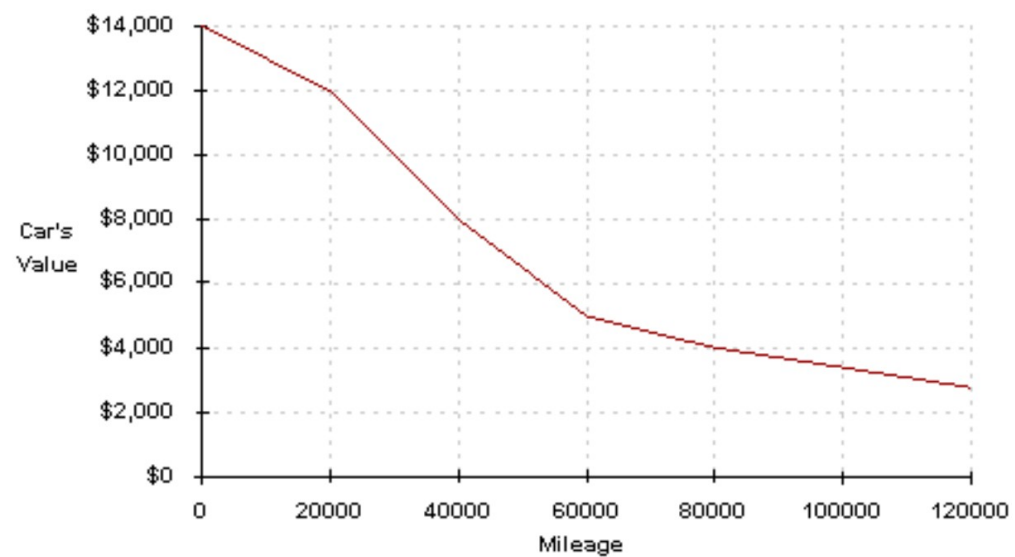
Example 1



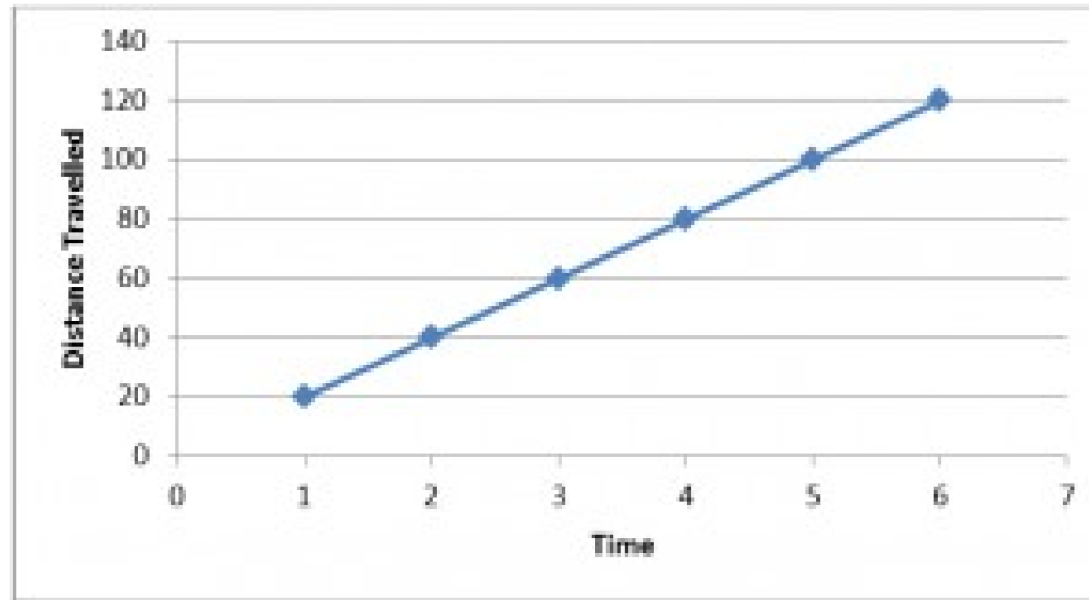
Example 2



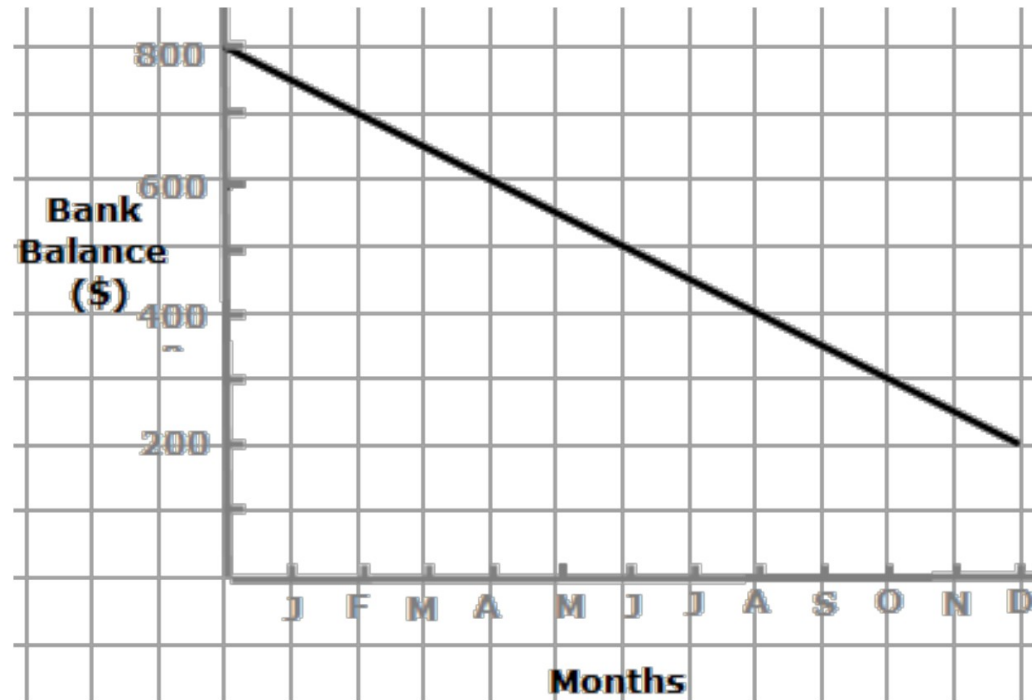
Example 3



Example 4



Example 5



PART 5

Making Your Own Graph



Six Steps to a Great Graph

1

Make a data table

Two labelled columns.

2

Choose graph type

Scatter / bar / pie?

3

Plot the data

Don't connect the dots.

4

Add a trendline

If linear or exponential.

5

Title the graph

'DV vs. IV'

6

Label the axes

Include units + scale.

Data Table Format

Make a data table

- Put your data in two x-y columns and label it

<i>x</i>	<i>y</i>
-2	-8
-1	-6.5
0	-5
1	-3.5
2	-2

MATH CLASS

Depth (km)	Basalt melting temperature (°C)
0	1100
25	1160
50	1250
100	1400
150	1600

SCIENCE CLASS

ONLY WRITE UNITS IN TOP BOXES WITH LABELS

Labeling Your Scatter Plot



The Title

Describe the RELATIONSHIP:

"The relationship between DV (y-axis) and IV (x-axis)"

Example: 'The relationship between distance travelled and time.'



The Axes

Each axis needs THREE things:

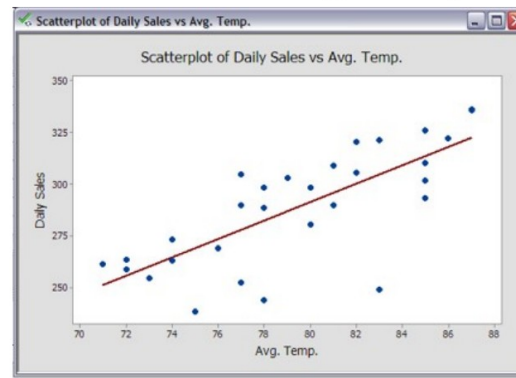
- WHAT (temperature, time, mass...)
- UNITS (°C, seconds, grams)
- SCALE (the numbers)

Missing any one = ambiguous graph.

Add a Trendline

Add a Trendline

- DO NOT CONNECT THE DOTS
- Add a trendline, linear, exponential or power series as appropriate
- A trendline is the average of your data



Labeling Bar & Pie Charts



The Title

Describe what's being compared.

Do NOT call it a relationship.

Use the word 'Comparison' — e.g. 'Comparison of energy sources in 2024.'



The Chart

For Bar/Column: label the axes.

For Pie: label each slice with its % share.

If you use color to encode categories, add a KEY.

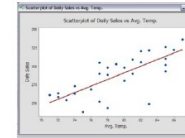
PART 6

Try It — Google Sheets

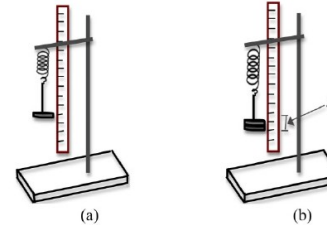


Practice 1 — Spring Force

Let's Plot a Relationship!



- The **Force (F)** a spring exerts is directly proportional to the **distance it is stretched** and can be written in the form of a linear $[y = f(x)]$ equation as $F = kx$, where k is the spring constant. Let's make a scatterplot of **$F = kx$** for **$x = 0, .025, .045, .080 \text{ \& } .095 \text{ m}$** where k equals 100.
- **$F = 100x$**



Practice 2 — Class Birthdays

Let's Make a Pie Chart!



- What percentage of students in this class have birthdays during the following quarters?
- Q1 Jan-Mar- 25%
- Q2 Apr-Jun-10%
- Q3 Jul-Sep- 5%
- Q4 Oct-Dec-60%

Practice 3 — Cars at the Rental

Let's Make a Bar/Column Graph



- Compare the different cars at the car rental agency when the following cars are available for rent:
- 10 Jeeps, 25 SUVs, 15 Economy Cars, 25 mid-size cars, 5 luxury cars, and 15 convertibles.

H O M E W O R K

Data Analysis Lab

Posted on Google Classroom. Due next class meeting.

Don't forget to click TURN IN.